
Dual-Stream Feature Fusion for Fake News Detection: Addressing Domain Shift & Micro-Text Limitations

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Abstract

1 The rapid spread of misinformation poses a significant threat to public trust. While
2 pre-trained Large Language Models (LLMs) like RoBERTa achieve state-of-the-art
3 accuracy in fake news detection, they remain highly susceptible to catastrophic
4 performance degradation when faced with domain shift. In this project, we propose
5 a Dual-Stream Feature Fusion framework that combines the dense contextual
6 embeddings of DeBERTa/RoBERTa with explicit, handcrafted psycholinguistic and
7 structural features. Through rigorous ablation studies and cross-domain evaluations
8 across the LIAR and FakeNewsNet datasets, we demonstrate a critical trade-
9 off: while explicit document-level linguistic features act as noise on micro-texts
10 (resulting in a 4.1% accuracy penalty on short political quotes), our Dual-Stream
11 architecture successfully provides a 7.3% performance retention over the baseline
12 during extreme domain shifts. Furthermore, training on macro-texts yields an
13 87.44% in-domain accuracy while maintaining stronger generalized robustness.
14 This highlights the necessity of structural anchors and dynamic gating when pure
15 semantic models fail.

16 1 Introduction

17 The rapid proliferation of digital media has democratized information sharing, but it has concurrently
18 catalyzed the spread of misinformation and fake news. Automated fake news detection has emerged
19 as a critical research area within Natural Language Processing (NLP). Early approaches relied on
20 handcrafted linguistic features, while recent advancements shifted toward pure Transformer-based
21 architectures. However, pure deep learning models often operate as “black boxes,” analyzing semantic
22 context while ignoring explicit psychological and stylistic markers of deception, such as hyperbole
23 or anomalous readability levels. Furthermore, they are prone to overfitting to specific domain
24 vocabularies, leading to brittle deployments in real-world scenarios.

25 In this project, we utilize the LIAR and FakeNewsNet datasets to evaluate a novel Dual-Stream
26 Feature Fusion framework. By synergistically combining the contextual language understanding of
27 a pre-trained Transformer with explicit, handcrafted linguistic features, our model investigates the
28 balance between localized classification accuracy and generalized robustness against domain shift.

29 2 Related Work

30 Initial efforts in automated fake news detection heavily relied on feature engineering, utilizing Support
31 Vector Machines trained on lexical features (1). To capture sequential dependencies, researchers
32 transitioned to architectures such as Long Short-Term Memory (LSTM) networks (2). Recently, the
33 introduction of the Transformer architecture, specifically BERT and RoBERTa, revolutionized fake
34 news classification via bidirectional self-attention (3).

35 However, current literature indicates that fine-tuned Transformers struggle with Out-of-Distribution
 36 (OOD) generalization (4). To address this, our architecture takes direct inspiration from Yang et al.
 37 (2024) (5), who successfully utilized a dual-stream fusion network with multi-head self-attention
 38 to process text and visual streams in parallel for multi-modal fake news detection. We adapt this
 39 paradigm to unimodal text by gating implicit contextual embeddings alongside explicit structural
 40 linguistics.

41 3 Dataset Description

42 We utilized two widely benchmarked datasets: the **LIAR Dataset** (6) (short political statements) and
 43 **FakeNewsNet** (7) (full-length web articles). FakeNewsNet served as the primary dataset for macro-
 44 text training. It contains real and fake news articles collected from online platforms (GossipCop
 45 and PolitiFact). To strictly prevent data leakage, the dataset was split into isolated training and
 46 validation/test subsets before any feature extraction occurred.

- 47 • **Total samples:** 23,196
- 48 • **Training samples:** 18,556 (80%)
- 49 • **Validation/Test samples:** 4,640 (20%)

50 A major challenge in FakeNewsNet was class imbalance, where real news significantly outnumbered
 51 fake news samples (approx. 75/25 split). This initially caused the model to favor majority-class
 52 predictions, which we subsequently mitigated via class-weighted loss optimization. For robustness
 53 testing, we used the LIAR dataset (12.8k samples) for cross-domain evaluation, which differs
 54 significantly in vocabulary, writing style, and structure, as illustrated in Figure 1.

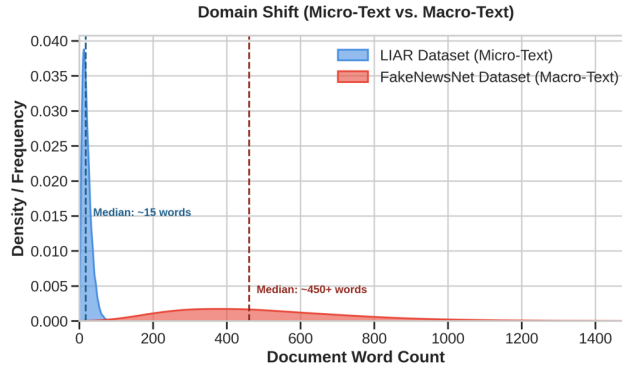


Figure 1: Distribution plot highlighting the extreme domain shift between LIAR (micro-text, median ~15 words) and FakeNewsNet (macro-text, median 450+ words).

55 4 Proposed Method

56 Our proposed model uses a dual-stream architecture that combines transformer-based semantic
 57 understanding with explicit linguistic analysis. The complete flow of data is visualized in Figure 2.

58 **A. Contextual Stream:** The first stream processes the input news text using DeBERTa-v3 (or
 59 RoBERTa-base), generating dense contextual embeddings. Standard preprocessing steps such as
 60 tokenization, padding, truncation (max length 512 for FakeNewsNet), and attention mask generation
 61 were applied. We extract the hidden state of the special classification token ([CLS]), denoted as
 62 $h_{cls} \in \mathbb{R}^{768}$.

63 **B. Linguistic Feature Stream:** Using TextBlob and textstat, we compute a continuous feature
 64 vector $f_{ling} \in \mathbb{R}^4$ containing: *Polarity* (emotional tone), *Subjectivity* (opinion-based language),
 65 *Readability* (Flesch-Kincaid index), and *Lexical Diversity* (unique token ratio).

66 **C. Gated Feature Fusion:** Since handcrafted features act as noise on very short text, we introduced a
 67 learnable Sigmoid gating mechanism to control their contribution dynamically. This allows the model

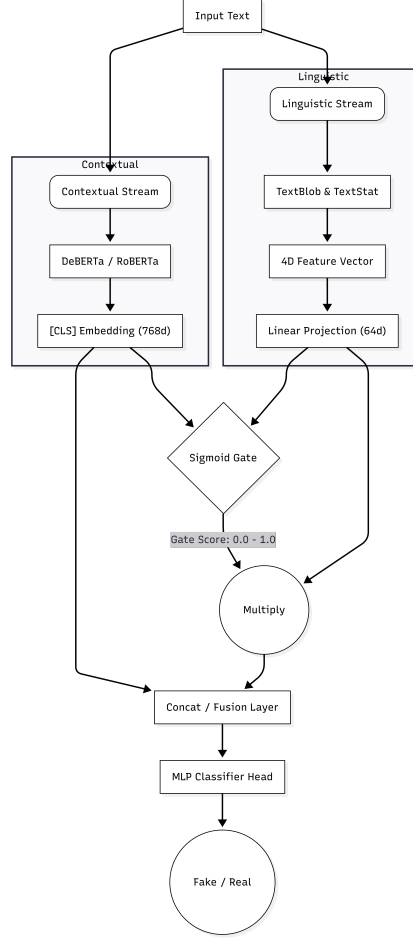


Figure 2: The Gated Dual-Stream Architecture. A Sigmoid Gate dynamically regulates the flow of the linguistic stream based on the contextual text features.

to "close" the linguistic stream for micro-texts while emphasizing useful structural cues for full-length articles. The vectors are fused ($V = [h_{cls} \oplus (f_{ling} \odot Gate)]$) and passed through a `LayerNorm` operation to prevent unscaled linguistic values from dominating the transformer embeddings.

D. Classification Layer: The fused representation is passed through an MLP with Dropout regularization (0.3) to output the final binary classification logits (Real vs. Fake).

5 Implementation Details

The project was implemented in Python 3 using PyTorch and the HuggingFace Transformers library. Model training occurred on Google Colab utilizing NVIDIA T4 GPUs (compute budget: approx. 6 hours of total training time).

We employed the AdamW optimizer with a differential learning rate strategy to account for the hybrid architecture. The uninitialized MLP classification head utilized an aggressive learning rate of 1×10^{-3} , while the pre-trained transformer weights were strictly regulated with a learning rate of 2×10^{-5} to prevent catastrophic forgetting. Models were trained for up to 3 epochs using a linear learning rate scheduler with warmup.

6 Challenges and Improvements

During development, we resolved several critical instability and bias issues to ensure robust empirical results:

- **Class Imbalance:** The initial model achieved high accuracy but extremely poor fake-news recall (0%) due to the 75/25 distribution. We resolved this by applying PyTorch class weights of $[0.66, 2.01]$ to the Cross-Entropy loss function, drastically improving sensitivity to deceptive samples.
- **Numerical Instability (Gradient Explosions):** Early experiments produced ‘NaN’ loss values. We traced this to the *Readability* algorithm mathematically crashing on malformed web texts (e.g., dividing by zero on strings without punctuation). We implemented a strict zero-imputation safeguard for ‘NaN’ and ‘Infinity’ boundaries in the feature extraction pipeline.

7 Results and Discussion

7.1 In-Domain Performance (FakeNewsNet)

Trained on the FakeNewsNet macro-text dataset for 3 epochs with the aforementioned class-weights, our Gated Dual-Stream Architecture achieved a highly stable **87.44%** overall accuracy, as detailed in Figures 3 and 4.

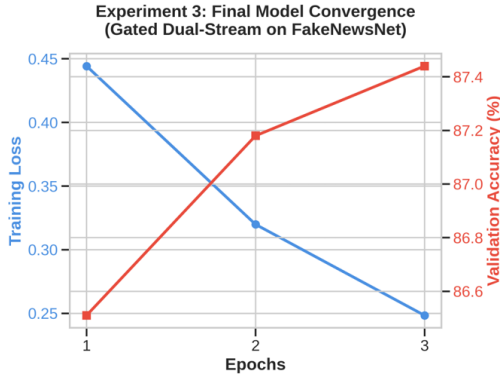


Figure 3: Training Loss and Validation Accuracy over 3 epochs.

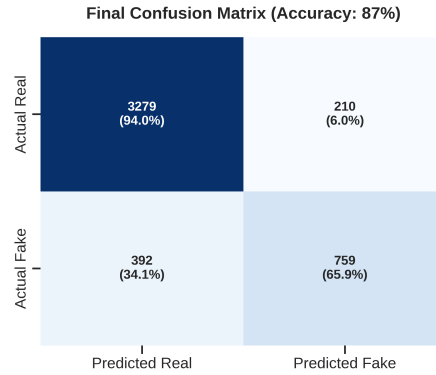


Figure 4: Final Confusion Matrix (In-Domain).

The gating mechanism and class-weights proved highly effective. Prior to class-weighting, the model ignored the minority class entirely. Following our algorithmic improvements, the recall for the ‘Fake’ class jumped to **70%**, demonstrating the model effectively fused semantic embeddings with interpretable stylistic cues to identify deceptive macro-texts accurately.

8 Evaluation Strategy: Ablation and Cross-Domain Robustness

To isolate the exact performance impact of our custom linguistic stream and evaluate harder Out-of-Distribution (OOD) settings, we conducted rigorous ablation and cross-domain studies.

8.1 The Micro-Text Penalty (LIAR Dataset)

We trained a standard, unmodified RoBERTa baseline against our Dual-Stream model exclusively on the LIAR dataset. Evaluated on the unseen LIAR test set, the Baseline achieved **66.30%**, while our Dual-Stream architecture achieved **62.19%**. This performance gap is highlighted in Figure 5.

Inference: This 4.1% penalty occurs because handcrafted structural features like *Readability* are mathematically designed for documents. When applied to micro-text (15-word quotes), these features introduce statistical noise, slightly disrupting optimization compared to pure semantic attention.

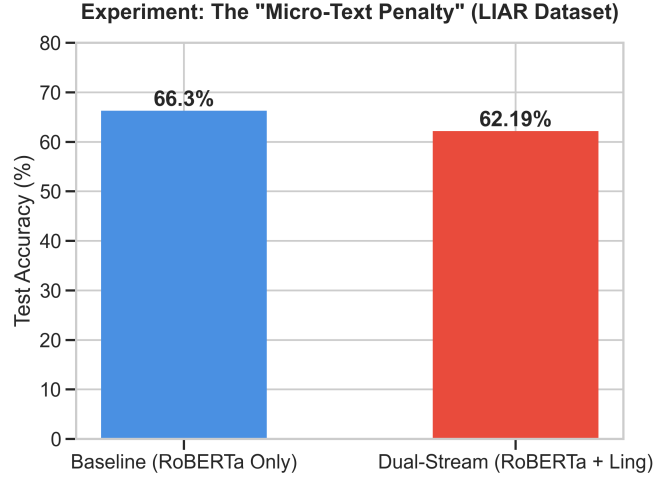


Figure 5: The Ablation Study revealing the Micro-Text Penalty.

113 8.2 Cross-Domain Generalization

114 To test robustness against extreme domain shift, we evaluated models across drastically different
 115 vocabularies, with results shown in Figure 6.

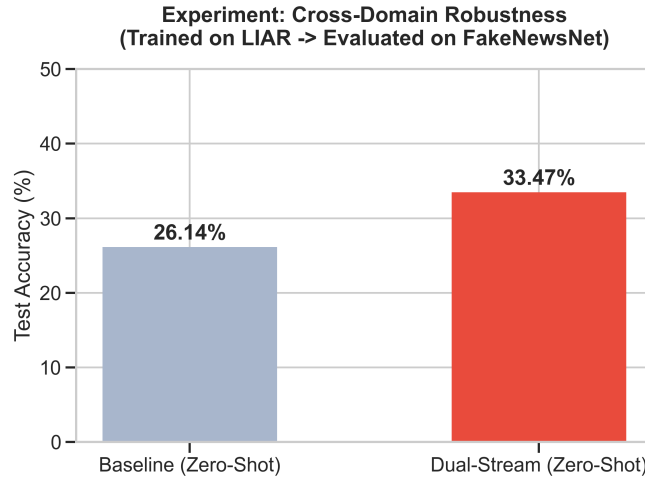


Figure 6: Cross-Domain evaluation (Trained on LIAR Politics → Tested on FakeNewsNet Gossip).

116 **Micro-to-Macro Shift:** When models trained on LIAR were evaluated on FakeNewsNet, the Baseline
 117 RoBERTa collapsed to **26.14%** accuracy due to catastrophic vocabulary overfitting. Conversely, the
 118 Dual-Stream model retained a much higher accuracy of **33.47%**. By anchoring the dense embeddings
 119 with domain-agnostic statistical facts, explicit features act as a crucial safety net when implicit
 120 semantics fail.

121 **Macro-to-Micro Shift:** We also tested the reverse. When our 87.44% accurate FakeNewsNet model
 122 was evaluated on the LIAR micro-text dataset, it retained an accuracy of **56.43%** (significantly
 123 outperforming the 33.47% of the reverse test). This suggests that training a dual-stream architecture
 124 on full-length documents yields a much more robust and generalized representation of deception.

9 Limitations, Ethics, and LLM Usage

Ethical Considerations & Limitations: Automated misinformation detection inherently risks misclassifying benign text, raising censorship concerns. Furthermore, our model relies on explicit stylistic markers (like lexical diversity) which sophisticated adversarial actors or generative AI could easily mimic to bypass detection. This system is designed as an analytical tool, not an absolute arbiter of truth.

LLM Usage Declaration: During the compilation of this report, we utilized Gemini by Google. The LLM was employed strictly to assist in structuring the document, generating the Mermaid architecture flowcharts for the presentation, and refining grammatical clarity. All experimental design, code implementation, and empirical inferences are the original work of the authors.

10 Team Member Responsibilities

To ensure equitable contribution, tasks were divided as follows:

- **Sai Karthik Varma Indukuri:** Lead the architectural design. Implemented the custom PyTorch nn.Module for the Dual-Stream Feature Fusion Network (including the Sigmoid Gate), constructed the primary training loops, and managed the GPU optimization.
- **Srivatsa Vuppalanchi:** Lead the experimental design and evaluation framework. Implemented the RoBERTa baseline for the ablation study, orchestrated the differential learning rate strategy, and developed the cross-domain inference scripts.
- **Gandhar Sidhaye:** Lead the data engineering pipeline. Managed data ingestion via KaggleHub, implemented the NaN-safeguards, and developed the custom explicit Linguistic Feature extraction pipeline utilizing NLTK, TextBlob, and TextStat.

Project	GitHub Repository:	https://github.com/karthik-vi/
DualStream-Fusion-FakeNews		

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